

tf.TensorArray Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/TensorArray

Class wrapping dynamic-sized, per-time-step, Tensor arrays.

Function Signatures

```
tf.TensorArray( dtype, size=None, dynamic_size=None,
clear_after_read=None, tensor_array_name=None, handle=None,
flow=None, infer_shape=True, element_shape=None,
colocate_with_first_write_call=True, name=None )
ta = tf.TensorArray(tf.float32, size=0, dynamic_size=True,
clear_after_read=False)
```

Code Examples

```
tf.TensorArray(  
dtype,  
size=None,  
dynamic_size=None,  
clear_after_read=None,  
tensor_array_name=None,  
handle=None,  
flow=None,  
infer_shape=True,  
element_shape=None,  
colocate_with_first_write_call=True,  
name=None  
)
```

```
tf.TensorArray(  
dtype,  
size=None,  
dynamic_size=None,  
clear_after_read=None,  
tensor_array_name=None,  
handle=None,  
flow=None,  
infer_shape=True,  
element_shape=None,  
colocate_with_first_write_call=True,  
name=None  
)
```

```
ta = tf.TensorArray(tf.float32, size=0, dynamic_size=True,  
clear_after_read=False)  
ta = ta.write(0, 10)  
ta = ta.write(1, 20)  
ta = ta.write(2, 30)  
ta.read(0)  
  
ta.read(1)  
  
ta.read(2)  
  
ta.stack()
```

```
ta = tf.TensorArray(tf.float32, size=0, dynamic_size=True,  
clear_after_read=False)
```

```
ta = ta.write(0, 10)
```

```
ta = ta.write(1, 20)
```

```
ta = ta.write(2, 30)
```

Documentation

Class wrapping dynamic-sized, per-time-step, Tensor arrays.

View aliases

Compat aliases for migration

See

Migration guide for

more details.

`tf.compat.v1.TensorArray`

Used in the notebooks

- Effective Tensorflow 2
 - Better performance with `tf.function`
 - Playing CartPole with the Actor-Critic method
 - Modeling COVID-19 spread in Europe and the effect of interventions
 - Neural machine translation with attention
 - Neural machine translation with a Transformer and Keras

This class is meant to be used with dynamic iteration primitives such as `while_loop` and `map_fn`. It supports gradient back-propagation via special "flow" control flow dependencies.

Note that although the array can be read multiple times and positions can be overwritten, behavior may be undefined when storing multiple references to the same array and `clear_after_read` is `False`. In particular, avoid using methods like `concat()` to convert an intermediate `TensorArray` to a `Tensor`, then further modifying the `TensorArray`, particularly if you need to backprop through it later.

Example 1: Plain reading and writing.

Example 2: Fibonacci sequence algorithm that writes in a loop then returns.

Example 3: A simple loop interacting with a `tf.Variable`.

Raises

Attributes

Methods

[View source](#)

Close the current TensorArray.

concat

[View source](#)

Return the values in the TensorArray as a concatenated Tensor.

All of the values must have been written, their ranks must match, and and their shapes must all match for all dimensions except the first.

gather

[View source](#)

Return selected values in the TensorArray as a packed Tensor.

All of selected values must have been written and their shapes must all match.

[View source](#)

identity

[View source](#)

Returns a TensorArray with the same content and properties.

[View source](#)

Read the value at location index in the TensorArray.

scatter

[View source](#)

Scatter the values of a Tensor in specific indices of a TensorArray.

[View source](#)

Return the size of the TensorArray.

[View source](#)

Split the values of a Tensor into the TensorArray.

[View source](#)

Return the values in the TensorArray as a stacked Tensor.

All of the values must have been written and their shapes must all match.

If input shapes have rank-R, then output shape will have rank-(R+1).

For example:

```
## unstack
```

[View source](#)

Unstack the values of a Tensor in the TensorArray.

If input value shapes have rank-R, then the output TensorArray will

contain elements whose shapes are rank-(R-1).

[View source](#)

Write value into index index of the TensorArray.